



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mrs. Mamatha
Student Name	Nehasree Padamati
Roll Number	2520030174
Branch	CSE
Course Outcome	CO3 – Graph Algorithms for Tourism Networks

Case Study Topic: Document Tracker – Document Workflow Connectivity Analysis using BFS, DFS and MST

Work Done Summary:

This case study models the Document Tracker system as a weighted undirected graph where nodes represent documents and edges represent dependencies or approval relationships between documents. Breadth-First Search (BFS) is used to identify all documents reachable from a selected document within a given number of levels. Depth-First Search (DFS) is used to explore document dependency chains and verify overall connectivity. Kruskal's algorithm computes the Minimum Spanning Tree (MST) to determine the minimum set of dependency links required to connect all documents efficiently.

Implementation Details:

1. Built an adjacency-list graph with 8 document nodes and 12 weighted dependency edges.
2. Ran BFS from "Doc-A" to find all documents reachable within 2 levels.
3. Ran DFS to check full connectivity and list all reachable documents in DFS order.
4. Applied Kruskal's algorithm (Union-Find) to compute MST – minimum dependency structure.
5. Verified MST total weight equals the sum of the smallest edges forming a spanning tree.
6. Visualised graph adjacency lists and MST edges in console output.

Result: Screenshots / Output

```
=== Document Workflow Analysis ===  
  
BFS Traversal:  
Doc-A  
Doc-B  
Doc-C  
Doc-D  
Doc-E  
  
DFS Traversal:  
Doc-A  
Doc-B  
Doc-F  
Doc-D  
Doc-E  
Doc-C  
Doc-G  
Doc-H  
  
MST Cost = 134  
  
Execution Successful  
  
=== Code Execution Successful ===
```

BFS from Doc-A (depth ≤ 2):
[Doc-A, Doc-B, Doc-C, Doc-D, Doc-E]

DFS complete order:
[Doc-A, Doc-B, Doc-F, Doc-D, Doc-E, Doc-C, Doc-G, Doc-H]

MST total cost: 134 units covering all 8 documents.

Time Complexity:

- BFS / DFS: $O(V + E)$
- Kruskal's MST: $O(E \log E)$

Graph-based tourism network enables scalable route planning and connectivity checks.

GitHub Link: https://github.com/nehasree2507-hash/DSA2-CaseStudies_Sem3

Student Signature	Faculty Signature